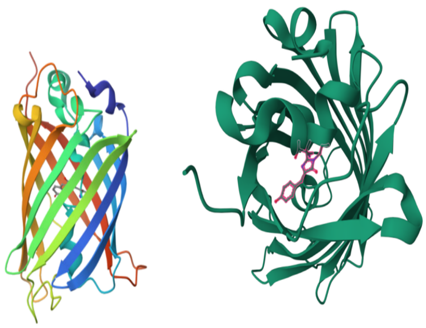
2026 Protein Design in Synbio challenges

1. 目标、规则、提交内容

本赛季基准对照升级为 **sfGFP**。赛道旨在挑战您运用计算生物学、生物信息学以及可编排的 Agent 工作流，在**全序列放开**的条件下，设计出兼具极高初始亮度与极限热稳定性的新型绿色荧光蛋白（GFP）“六边形战士”

Superfolder Green Fluorescent Protein (sfGFP)



1.1 您的提交内容（核心序列文件、说明文档、开源仓库代码）

参赛队伍需在 SynBio Challenges 官网严格按以下格式提交三项内容：

☐ 设计序列文件 (.csv)

○ **数量与长度**：每队最多提交 **6 条** 自己设计的蛋白序列。序列长度严格限制在 220 - 250 aa 之间（没有以往位点数量的限制，本次为全序列自由设计）。

○ **字符规范**：序列**必须以蛋氨酸（M）开头**。仅允许使用 20 种标准氨基酸的大写字母，严禁包含终止密码子（如 *）或标点符号。

○ 本次由大设施统一合成 DNA 模板（已含启动子、RBS 及终止子，且浓度标准化），因此**您只需提供氨基酸序列**。本赛事使用开源 DNA Chisel 算法执行反向翻译。

○ 请在提交前逐条与 Exclusion_List.csv 中的序列做完全一致性比对，确保您提交的 6 条序列均未出现在我们提供的 Exclusion_List.csv 文件中。

○ **表头规范**：共三列，必须包含 Team_Name（队伍名），Seq_ID（如1,2,3...），Sequence (氨基酸序列)，可参考 submission_template.csv

A	B	C
Team_Name	Seq_ID	Sequence
YourTeamName	1	M......(REPLACE_WITH_YOUR_DESIGNED_SEQUENCE_220TO250AA)
YourTeamName	2	M......(REPLACE_WITH_YOUR_DESIGNED_SEQUENCE_220TO250AA)
YourTeamName	3	M......(REPLACE_WITH_YOUR_DESIGNED_SEQUENCE_220TO250AA)
YourTeamName	4	M......(REPLACE_WITH_YOUR_DESIGNED_SEQUENCE_220TO250AA)
YourTeamName	5	M......(REPLACE_WITH_YOUR_DESIGNED_SEQUENCE_220TO250AA)
YourTeamName	6	M......(REPLACE_WITH_YOUR_DESIGNED_SEQUENCE_220TO250AA)

☐ **设计思路与说明文档（.pdf)**

○ 展示从零开始或组合现有模型（如 ESM, ProteinMPNN 等）生成序列的完整流程。阐述您的算法管线、如何平衡亮度和稳定性目标，以及最终筛选这 6 条序列的原因。

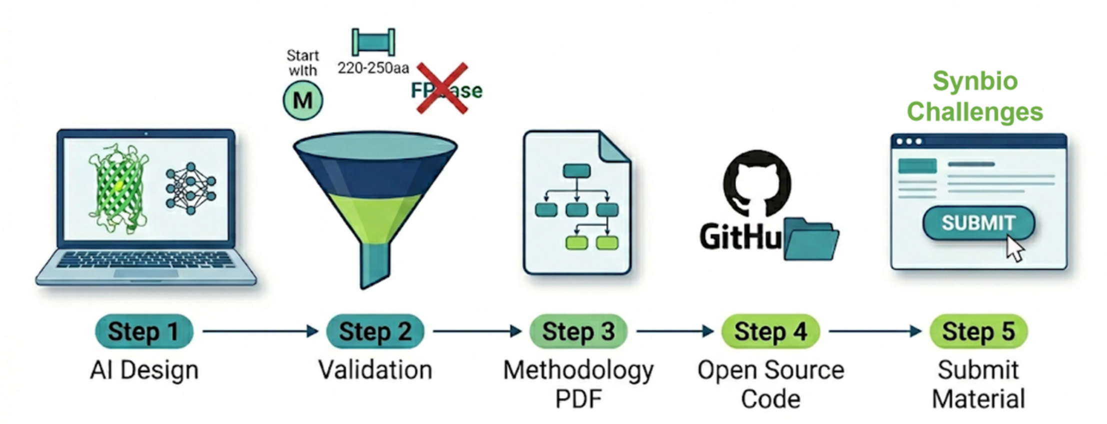
○ LLM **Agent**：鼓励并欢迎使用各种大模型与代码 Agent。若使用智能体设计，需在文档中展示 Agent 的逻辑树和关键执行日志，以确保过程可复现。

○ 该文档将作为评审团队的展示与思路参考。在客观得分相近的情况下，优先选择思路清晰、技术路径前沿且开源的队伍进行宣传和展示。

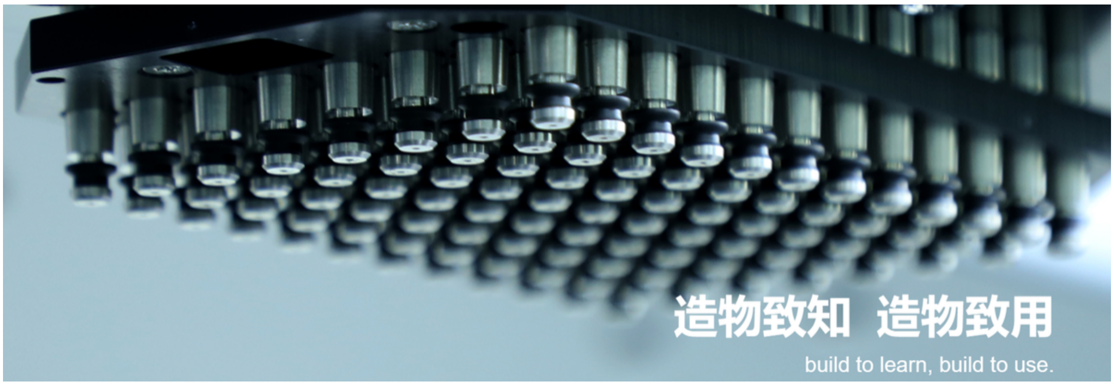
☐ **开源项目链接（URL)**

○ 须提交公开的 GitHub / GitLab / HuggingFace 等代码托管仓库链接，确保其它队伍可以复现。

○ **根目录需包含 README.md**，明确说明环境配置要求、模型依赖库以及推理代码运行方式。



1.2 自动化实验流程 (由大设施执行)



所有序列的测试由大设施统一执行“**一次表达 + 两次读数**”流水线。本次实验采用的无细胞体系为 **NEBexpress Cell-free E. coli Protein Synthesis System**。

- **步骤一：初始亮度检测**

1. **表达**：在黑色平底 96 孔板中加入 Cell-Free 反应液与标准化浓度的 DNA 模板。置于恒温振荡器，**30°C** 孵育 3 小时（保持振荡供氧）。

2. **成熟**：降温至 **25°C** 静置 30 分钟，确保发色团完全成熟。

3. **读数 1**：使用酶标仪读取初始荧光强度（Ex: 485nm / Em: 515nm），记为您序列的 $F_{initial}$。同批次野生型 sfGFP 的亮度记为 $F_{initialWT}$。

- **步骤二：极端热稳定性检测**

1. **加热**：自动化工作站将无细胞表达产物转移至 PCR 板并封膜。在 PCR 仪中执行 **72°C** 高温加热 **10 分钟**。

2. **复性**：PCR 仪自动降温至 **25°C** 并保持 **5 分钟**（消除温度的物理淬灭效应，仅测试因变性导致的不可逆荧光丧失）。

3. **读数 2**：读取加热后的残余荧光强度，记为F_{final}。

1.3 得分规则

本次赛事的评分采用“双能乘积制”，要求挑战者拒绝偏科，追求极限。

- **项一：相对亮度得分** —— 评估蛋白在 Cell-Free 体系中的折叠效率与消光系数。

$$\text{相对亮度得分} = F_{initial} \ / \ F_{initialWT}$$

- **项二：热稳定性保留率** —— 评估蛋白抵抗 72°C 高温变性的能力。

$$\text{热稳定性得分} = F_{final} \ / \ F_{initial}$$

- **极低亮度阈值淘汰**

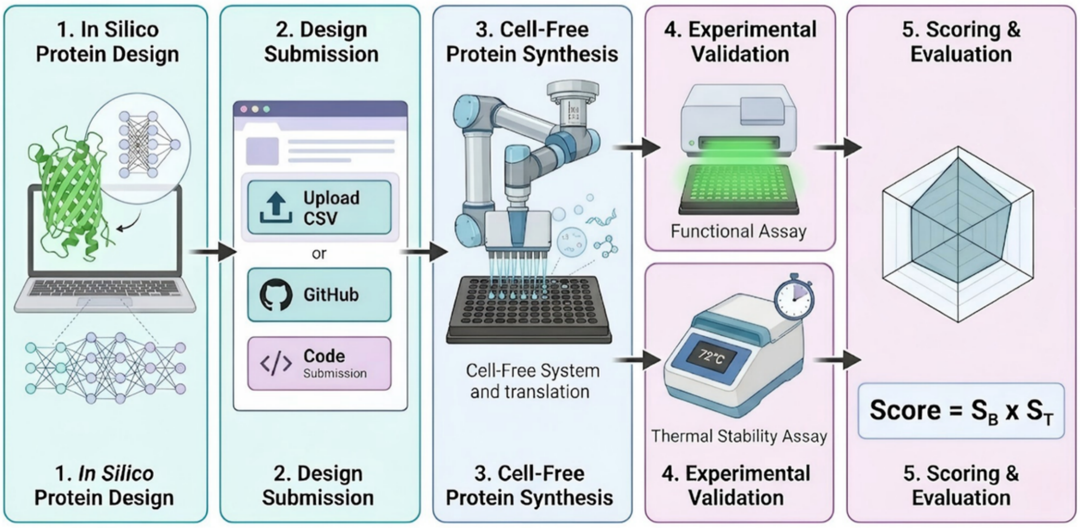
若某条提交序列的初始亮度 **不到同批次野生型 sfGFP 的 30%**（即

$F_{initial} < 0.3 \times F_{initialWT}$)，说明该序列基础结构/功能遭到严重破坏，该序列直接淘汰出局，此条序列两项及总成绩均记为 **0 分**。

- **双能综合得分**

$$\text{单条序列综合得分} = \text{相对亮度得分} \times \text{热稳定性保留率} = \frac{F_{initial}}{F_{initialWT}} \times \frac{F_{final}}{F_{initial}}$$

- **排名与奖项分配**
  - **金/银奖**：我们将以您提交的 6 条序列中**得分最高的一条(Best Top-1)**作为本队的最终打榜成绩进行排榜。前 30% 金奖、接下来若干比例银奖。
  - 此外，设立由单项分决出的**最佳亮度奖**和**最佳热稳定奖**作为独立荣誉。



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2. 数据与资料介绍

2.1 GFP 突变与亮度数据

提供涵盖历史多品种 GFP 蛋白的数据集（GFP_data.xlsx），可用作大模型微调训练、序列-功能关系的预训练。

(注：原始数据虽多为历史 avGFP/amacGFP 等短变体，并不直接等同于本赛季的 sfGFP 提交空间，但掌握其中的普遍折叠密码和发色团微环境可很好地迁移至全序列设计)

2.2 往年的 Top 序列

GFP_data.xlsx 中 sheet-beforetopseqs 提供历年优秀序列共 20 条（2024、2025 各 10 条）及对应 year、sequence 字段，供您追溯历史特征工程设计思路。

2.3 蛋白参考序列与蛋白质结构 PDB 推荐

数据包提供 sfGFP、avGFP、amacGFP、cgreGFP、ppluGFP 共五条参考氨基酸序列及推荐 PDB（见 AAseqs of 5 GFP proteins.txt），供选手构建三维图神经网络（GNN）、Structure-based Agent 或逆向折叠模型（ProteinMPNN）时使用

> *>sfGFP*
>
> *MSKGEELFTGVVPILVELDGDVNGHKFSVRGEGEGDATNGKLTLKFICTTGKLPVPWPTLVTTLTYGVQCFSRYPD*
>
> *HMKRHDFFKSAMPEGYVQERTISFKDDGTYKTRAEVKFEGDTLVNRIELKGIDFKEDGNILGHKLEYNFNSHNVYIT*
>
> *ADKQKNGIKANFKIRHNIVEDGSVQLADHYQQNTPIGDGPVLLPDNHYLSTQSVLSKDPNEKRDHMVLLEFVTAAGI*
>
> *THGMDELYK*
>
> *# recommend PDB: 2B3P*

2.4 排除序列列表

Exclusion_List.csv。此文件包含禁止重复提交的序列名单（包含历史测过的序列、基于 FPbase 的天然或已知变体等），任何与排除列表中任意一条序列完全一致的提交序列视为无效，另可参考 FPbase(https://www.fpbase.org/ ）等公开资源避免重复已知全长变体。

Exclusion_List

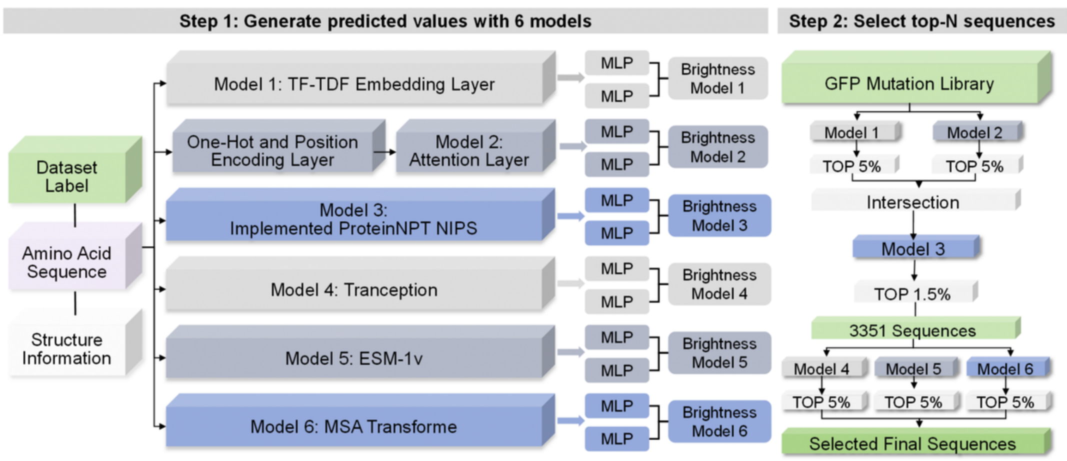
2.5 热稳定性资料

由于缺乏针对 72°C 温度直接测量的庞大开源数据集，建议选手多方拓展：

- **先验知识引入**：参考 sfGFP 本身（自带多重折叠极性突变）的基础，探索在β桶状结构域改进表面电荷网络、计算填补核心疏水空腔（如 Rosetta Pack 功能）等先验逻辑。
- **模型降维**：利用具有物理或结构感知能力的语言模型（如 ESM-3，SaProt）对耐热域嵌入处理进行对比分析学习，提取耐热特征。

2.6 往届优秀队伍思路参考

基于异构 AI 模型集成与多级级联漏斗筛选的 GFP 高荧光 Fitness 框架在特征学习阶段，算法并行构建了 6 个涵盖不同归纳偏置的异构模型（从 TF-IDF 离散表征、拓扑注意力机制，到 ESM-1v、Tranception 等前沿的大型蛋白质语言模型），实现序列及进化特征到荧光强度的定量回归预测；随后，对高通量 GFP 突变文库实施梯队式降维过滤：初筛阶段利用轻量级模型（Model 1 & 2）截留 Top 5%预测结果并取交集以快速收敛搜索空间；中级过滤阶段引入高分辨率模型（Model 3）将候选池进一步提纯至 Top 1.5%（约 3351 条序列）；终筛阶段则交由三大复杂预训练模型（Model 4/5/6）执行交叉验证与联合阈值截断



2.7 竞赛入门教程

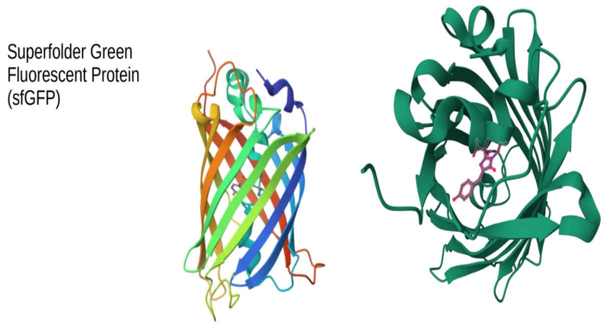
不会部署模型？不知道从哪里开始？您可以参考访问基础教程教程为 2025 思路参考，2026 须自行改路径、列名、WT、长度与提交格式：

https://www.bohrium.com/notebooks/36787332597

2026 Protein Design in SynBio Challenges(english)

1. Objectives, Rules, and Submission Content

This season, the baseline control has been upgraded to **sfGFP**. This track challenges you to utilize computational biology, bioinformatics, and orchestratable Agent workflows to design a novel Green Fluorescent Protein (GFP) **"an all-around champion"** that possesses both extremely high initial brightness and extreme thermal stability, under the condition of full-sequence open design.



1.1 Your Submission Content

Participating teams must submit three items on the official SynBio Challenges website, strictly following the formats below:

☐ Design Sequence File (.csv)

- **Quantity & Length:** Each team may submit a maximum of **6** self-designed protein sequences. The sequence length is strictly limited to between **220 - 250 aa** (There are no limitations on the number of mutation sites as in previous years; this time it is a full-sequence free design).

- **Character Norms:** Sequences must start with Methionine (M). Only the uppercase letters of the 20 standard amino acids are allowed. Stop codons (e.g., *) or punctuation marks are strictly prohibited.

- **Synthesis:** The mega-facility will uniformly synthesize the DNA templates (already containing the promoter, RBS, and terminator, with standardized concentration), so you only need to provide the amino acid sequences.Reverse translation will be performed using the open-source DNA Chisel algorithm.

- **Exclusion Check:** Before submitting, please cross-check each sequence against the Exclusion_List.csv to ensure that none of your 6 submitted sequences appear in the provided exclusion list.

- **Header Norms:** The file must contain exactly three columns with the headers: Team_Name, Seq_ID (e.g., 1, 2, 3…), and Sequence (amino acid sequence). Please refer to submission_template.csv.

A	B	C
Team_Name	Seq_ID	Sequence
YourTeamName	1	M...... (REPLACE_WITH_YOUR_DESIGNED_SEQUENCE_220TO250AA)
YourTeamName	2	M...... (REPLACE_WITH_YOUR_DESIGNED_SEQUENCE_220TO250AA)
YourTeamName	3	M...... (REPLACE_WITH_YOUR_DESIGNED_SEQUENCE_220TO250AA)
YourTeamName	4	M...... (REPLACE_WITH_YOUR_DESIGNED_SEQUENCE_220TO250AA)
YourTeamName	5	M...... (REPLACE_WITH_YOUR_DESIGNED_SEQUENCE_220TO250AA)
YourTeamName	6	M...... (REPLACE_WITH_YOUR_DESIGNED_SEQUENCE_220TO250AA)

☐ Design Concept and Documentation (.pdf)

- **Workflow:** Showcase your complete pipeline for generating sequences, whether from scratch or by combining existing models (e.g., ESM, ProteinMPNN, etc.). Explain your algorithm pipeline, how you balanced the brightness and stability objectives, and the reasoning behind selecting these final 6 sequences.

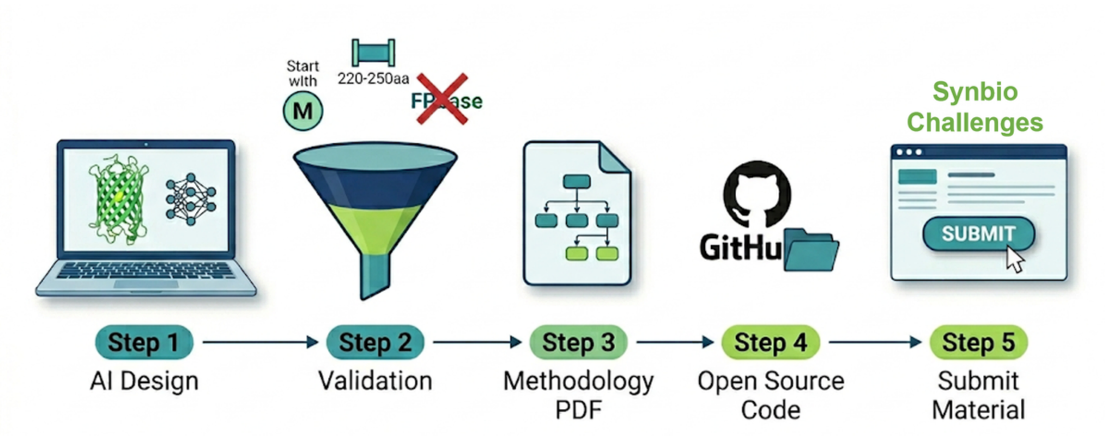
- **LLM Agent:** We encourage and welcome the use of various Large Language Models and code Agents. If an Agent was used for the design, you must display the Agent's logic tree and key execution logs in the document to ensure the process is reproducible.

- **Evaluation:** This document will serve as a presentation and conceptual reference for the judging panel. In cases where objective scores are similar, teams with clear logic, cutting-edge technical pathways, and open-source practices will be prioritized for promotion and showcase.

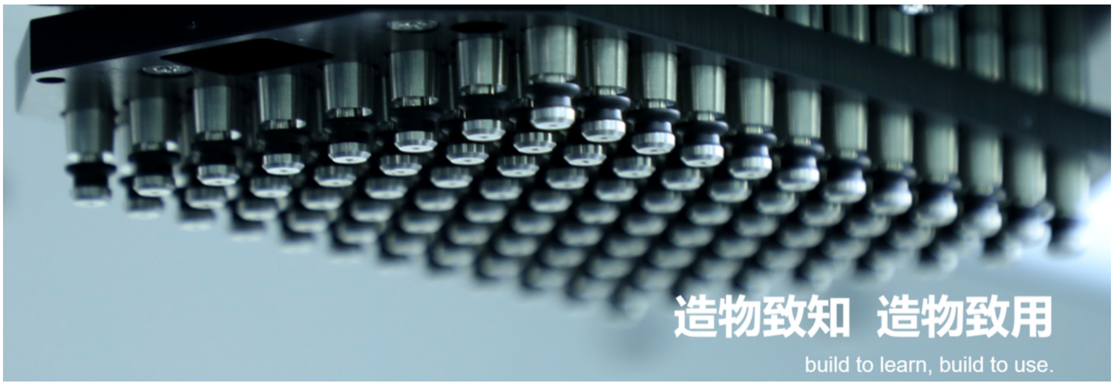
☐ Open-Source Project Link (URL)

- You must submit a public code hosting repository link (GitHub / GitLab / HuggingFace...) to ensure other teams can reproduce your work.

- The root directory must contain a README.md that clearly explains the environment configuration requirements, model dependency libraries, and instructions on how to run the inference code.



1.2 Automated Experimental Workflow



Step 1: Initial Brightness Detection

1. **Expression:** Incubate DNA templates in a cell-free reaction at 30°C for 3 hours.
2. **Maturation:** Cool to 25°C for 30 minutes for chromophore maturation.
3. **Reading 1:** Measure initial fluorescence (Ex: 485nm / Em: 515nm). Your sequence's brightness:$F_{initial}$
   a. Wild-type sfGFP brightness: $F_{initialWT}$

Step 2: Extreme Thermal Stability Detection

1. **Heating:** Heat the products at **72°C for 10 minutes** in a PCR machine.
2. **Renaturation:** Cool to 25°C for 5 minutes to eliminate physical quenching.
3. **Reading 2:** Measure residual fluorescence intensity. Residual brightness: F_{final}

1.3 Scoring Rules

The scoring adopts a "**Dual-Performance Product System,**" requiring challengers to pursue extreme performance in both categories.

- **Metric 1: Relative Brightness Score**

Evaluates folding efficiency and extinction coefficient in the Cell-Free system.

$$\text{Relative Brightness Score} = \frac{F_{initial}}{F_{initialWT}}$$

- **Metric 2: Thermal Stability Retention Rate**

Evaluates the ability to resist denaturation at 72°C.

$$\text{Thermal Stability Retention Score} = \frac{F_{final}}{F_{initial}}$$

- **Extremely Low Brightness Elimination**

If a sequence's initial brightness is **less than 30%** of the wild-type sfGFP:

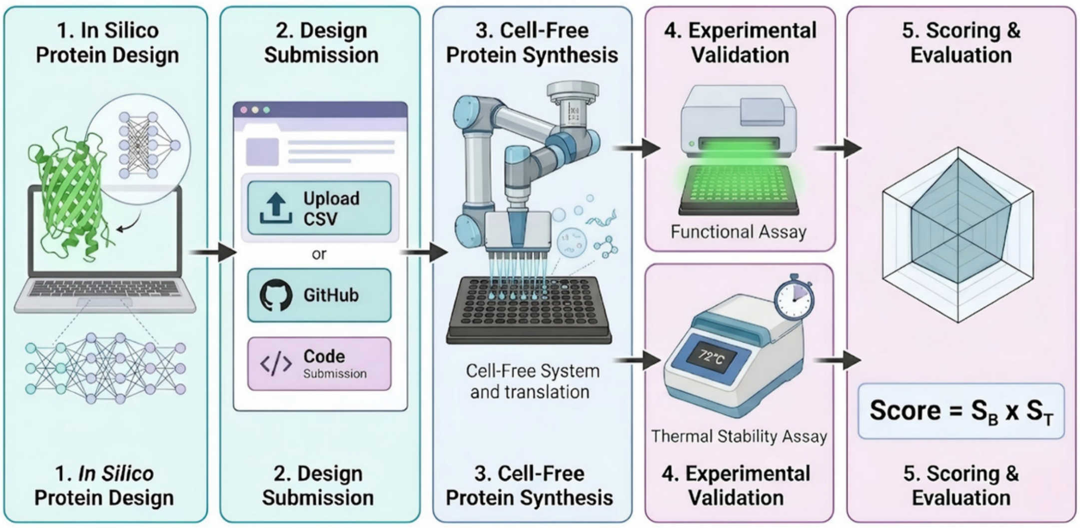
$$F_{initial} < 0.3 \times F_{initialWT}$$

The sequence is **disqualified**, and its scores are recorded as **0**.

- **Comprehensive Dual-Performance Score**

$$\text{Total Score} = \text{Relative Brightness Score} \times \text{Thermal Stability Retention Rate} = \frac{F_{initial}}{F_{initialWT}} \times \frac{F_{final}}{F_{initial}}$$

- **Ranking and Awards**
  - **Gold/Silver Awards:** Based on your team's **Best Top-1** sequence score. Top 30% receive Gold Awards.
  - **Special Honors:** Independent awards for "Best Brightness" and "Best Thermal Stability."



2. Data and Resources Introduction

2.1 GFP Mutation and Brightness Data

We provide a dataset (GFP_data.xlsx) covering historical data of multiple GFP protein variants, which can be used for LLM fine-tuning and pre-training of sequence-function relationships.

(Note: Although the raw data mostly consists of historical short variants like avGFP/amacGFP and does not directly equate to this season's sfGFP submission space, mastering the universal folding codes and chromophore microenvironments within them can be well transferred to full-sequence design).

2.2 Top Sequences from Previous Years

The **beforetopseqs worksheet** in **GFP_data.xlsx** provides a total of 20 outstanding sequences from previous years along with corresponding year and sequence fields, allowing you to trace historical feature engineering design concepts.

2.3 Reference Protein Sequences and Recommended PDBs

The data package provides five reference amino acid sequences (sfGFP, avGFP, amacGFP, cgreGFP, ppluGFP) and recommended PDBs (see AAseqs of 5 GFP proteins.txt). These are provided for participants to use when building 3D Graph Neural Networks (GNNs), Structure-based Agents, or inverse folding models (like ProteinMPNN).

> >sfGFP
>
> MSKGEELFTGVVPILVELDGDVNGHKFSVRGEGEGDATNGKLTLKFICTTGKLPVPWPTLVTTLTYG
>
> VQCFSRYPDHMKRHDFFKSAMPEGYVQERTISFKDDGTYKTRAEVKFEGDTLVNRIELKGIDFKEDG
>
> NILGHKLEYNFNSHNVYITADKQKNGIKANFKIRHNIVEDGSVQLADHYQQNTPIGDGPVLLPDNHYL
>
> STQSVLSKDPNEKRDHMVLLEFVTAAGITHGMDELYK
>
> # recommend PDB: 2B3P

2.4 Exclusion List

Exclusion_List.csv. This file contains a list of sequences that are prohibited from being submitted (including historically tested sequences, natural or known variants based on FPbase, etc.). Any submitted sequence that is completely identical to any sequence in the exclusion list will be considered invalid. You may also refer to public resources like FPbase(https://www.fpbase.org/) to avoid duplicating known full-length variants.

Exclusion_List

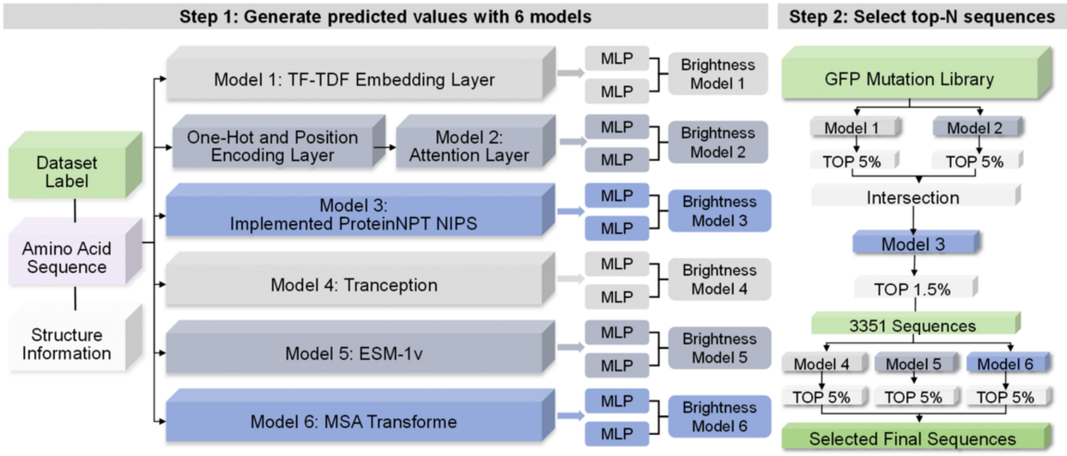
2.5 Thermal Stability Resources

Due to the lack of massive open-source datasets directly measured at 72°C, participants are advised to expand their approaches in multiple ways:

- **Introduction of Prior Knowledge:** Based on sfGFP itself (which inherently has multiple folding polarity mutations), explore prior logic such as introducing disulfide bond combinations improving the surface charge network, and computationally filling the core hydrophobic cavity (e.g., Rosetta Pack function).
- **Model Dimensionality Reduction:** Utilize language models with physical or structural awareness (e.g., ESM-3, SaProt) to conduct comparative analysis and learning on heat-resistant domain embeddings to extract heat-resistant features.

2.6 Reference for Previous Outstanding Teams’ Approaches

Example: A high-fluorescence Fitness framework for GFP based on heterogeneous AI model integration and multi-stage cascade funnel screening. In the feature learning stage, the algorithm built 6 heterogeneous models in parallel covering different inductive biases (from TF-IDF discrete representations and topological attention mechanisms to cutting-edge large protein language models like ESM-1v and Tranception) to achieve quantitative regression prediction of fluorescence intensity from sequence and evolutionary features. Subsequently, a tiered dimensionality reduction filtration was applied to the high-throughput GFP mutation library: the initial screening stage used lightweight models (Model 1 & 2) to intercept the Top 5% prediction results and take the intersection to quickly converge the search space; the intermediate filtration stage introduced a high-resolution model (Model 3) to further purify the candidate pool to the Top 1.5% (about 3,351 sequences); the final screening stage was handed over to three major complex pre-trained models (Model 4/5/6) to execute cross-validation and joint threshold truncation.



2.7 Competition Starter Tutorial

Struggling with model deployment? Not sure where to start? You can refer to the basic tutorials for 2025 as a reference; for 2026, you will need to modify the paths, column names, WT, lengths, and submission formats on your own.

https://www.bohrium.com/notebooks/36787332597

Wishing you the best of luck in creating a “perfect protein” that breaks the rules of life in 2026!

3.参考文献/Reference

[1] mBaoJin.pdf

> Zhang H, Lesnov GD, Subach OM, Zhang W, Kuzmicheva TP, Vlaskina AV, Samygina VR, Chen L, Ye X, Nikolaeva AY, Gabdulkhakov A, Papadaki S, Qin W, Borshchevskiy V, Perfilov MM, Gavrikov AS, Drobizhev M, Mishin AS, Piatkevich KD, Subach FV. Bright and stable monomeric green fluorescent protein derived from StayGold. Nat Methods 2024; 21: 657–665. https://doi.org/10.1038/s41592-024-02203-y

[2] nature-Local fitness landscape of the green fluorescent protein.pdf

> Sarkisyan KS, Bolotin DA, Meer MV, Usmanova DR, Mishin AS, Sharonov GV, Ivankov DN, Bozhanova NG, Baranov MS, Soylemez O, Bogatyreva NS, Vlasov PK, Egorov ES, Logacheva MD, Kondrashov AS, Chudakov DM, Putintseva EV, Mamedov IZ, Tawfik DS, Lukyanov KA, Kondrashov FA. Local fitness landscape of the green fluorescent protein. Nature 2016; 533: 397–401. https://doi.org/10.1038/nature17995

[3] Staygold.pdf

> Hirano M, Ando R, Shimozono S, Sato H, Kaneko R, Okada Y, Niino Y, Miyawaki A. A highly photostable and bright green fluorescent protein. Nat Biotechnol 2022;40:1132–1142. https://doi.org/10.1038/s41587-022-01278-2

[4] superfolder.pdf

> Pedelacq J-D, Cabantous S, Tran T, Terwilliger TC, Waldo GS. Engineering and characterization of a superfolder green fluorescent protein. Nat Biotechnol 2006;24:79–88. https://doi.org/10.1038/nbt1172

[5] TGP, an extremely stable, non-aggregating fluorescent protein.pdf

> Close DW, Paul CD, Langan PS, Wilce MCJ, Traore DAK, Halfmann R, Rocha RC, Waldo GS, Payne RJ, Rucker JB, Prescott M, Bradbury ARM. Thermal green protein, an extremely stable, nonaggregating fluorescent protein created by structure-guided surface engineering. Proteins 2015; 83: 1225–1237. https://doi.org/10.1002/prot.24699